# IGCSE 0607 Practice Questions — E7.2 — Combined Questions & Markscheme

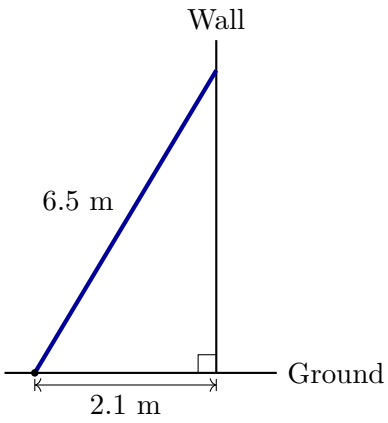
## Right-angled triangles

**Total Questions:** 6 — **Total Marks:** 48 Marks

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- **E7.2** Right-angled triangles
  → **Paper 4 (Calculator)** *(6 Qs)*

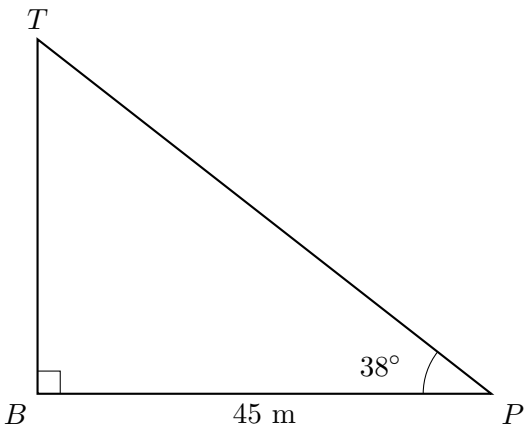
**1** A straight ladder of length 6.5 m leans against a vertical wall.
The base of the ladder is 2.1 m from the bottom of the wall on horizontal ground.




Calculate the angle the ladder makes with the horizontal ground.

..... ° **[3]**

**2** A surveyor stands at a point $P$, which is 45 m away from the base, $B$, of a vertical tower $TB$.
The angle of elevation of the top of the tower, $T$, from $P$ is 38°.




**(a)** Calculate the height of the tower, $TB$.

..... $m$ **[2]**

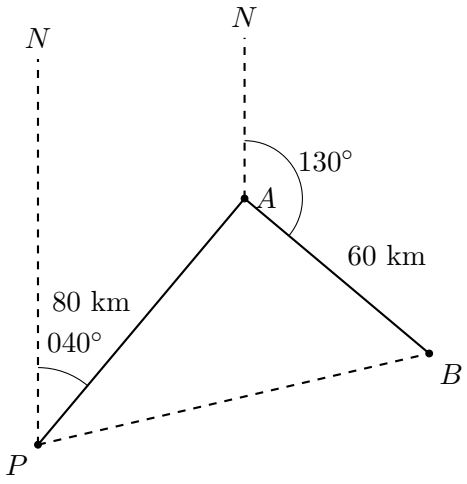
**(b)** Calculate the straight-line distance $PT$.

..... $m$ **[2]**

**(c)** A bird sits at a point $X$ on the tower, where $BX = 20$ m.
Calculate the angle of elevation of the bird from $P$.

..... ° **[3]**

**3** A ship sails from port $P$ on a bearing of 040° for 80 km to point $A$.
It then changes course and sails on a bearing of 130° for 60 km to point $B$.




**(a)** Calculate the direct distance from $P$ to $B$.

$PB =$ ..... km **[3]**

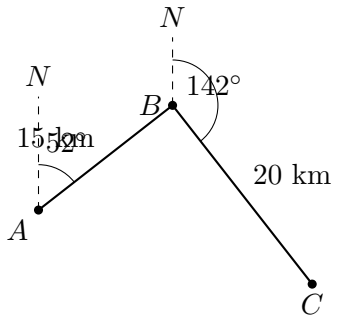
**(b)** Calculate the bearing of $B$ from $P$.

Bearing = ..... ° **[3]**

**(c)** Calculate the shortest distance from point $A$ to the path $PB$.

..... km **[3]**

**4** A ship sails from port $A$ to point $B$ on a bearing of 052° for 15 km.
It then sails from point $B$ to point $C$ on a bearing of 142° for 20 km.




**(a)** Show that angle $ABC = 90°$.

**[2]**

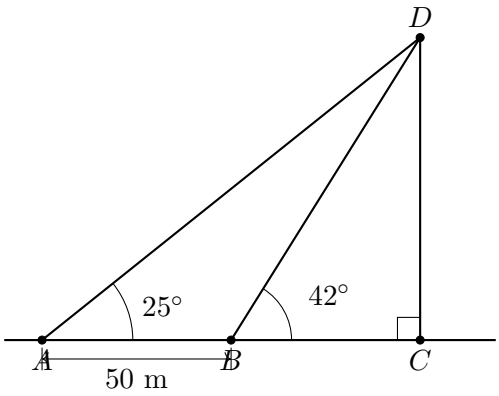
**(b)** Calculate the distance $AC$.

..... km **[3]**

**(c)** Calculate the bearing of $C$ from $A$.

..... ° **[4]**

**5** Point $A$ and point $B$ are on horizontal ground, in a straight line with the base $C$ of a vertical building $CD$.
The distance between $A$ and $B$ is 50 m.
The angle of elevation of the top of the building, $D$, from $A$ is 25° and from $B$ is 42°.



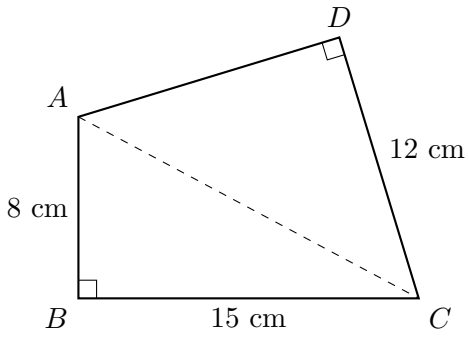

**(a)** Calculate the height of the building $CD$.

$CD =$ ..... $m$ **[5]**

**(b)** A bird flies directly along the straight path from $A$ to $D$.
Calculate the shortest distance from point $B$ to the flight path $AD$.

..... $m$ **[3]**

**6** The diagram shows a quadrilateral $ABCD$.
Angle $ABC = 90°$ and angle $ADC = 90°$.
$AB = 8$ cm, $BC = 15$ cm, and $CD = 12$ cm.




**(a)** Calculate the length of $AC$.

..... cm **[2]**

**(b)** Calculate the length of $AD$.

..... cm **[3]**

**(c)** Calculate the shortest distance from $D$ to the line $AC$.

..... cm **[3]**

**(d)** Calculate the size of angle $DAB$.

..... ° **[4]**

**1** Identifies cosine ratio: $\cos(\theta) = \dfrac{2.1}{6.5}$ **[M1]**

$\theta = \cos^{-1}\left(\dfrac{2.1}{6.5}\right)$ **[M1]**

71.2 **[A1]**

**[Back to Question]** *[Total Marks: 3]*

**2 (a)** $\tan(38°) = \dfrac{TB}{45}$ **[M1]**

35.2 (or 35.15 to 35.16) **[A1]**

**(b)** $\cos(38°) = \dfrac{45}{PT}$ or $PT^2 = 45^2 + 35.15^2$ **[M1]**

57.1 (or 57.10 to 57.11) **[A1]**

**(c)** $\tan(XPB) = \dfrac{20}{45}$ **[M1]**

Angle $= \tan^{-1}\left(\dfrac{20}{45}\right)$ **[M1]**

24.0 (or $23.96\ldots$) **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**3 (a)** Calculates $\angle PAB = 360° - (180° - 40°) - 130° = 90°$ (or similar valid reasoning to show right angle) **[M1]**

Uses Pythagoras' theorem: $PB^2 = 80^2 + 60^2$ **[M1]**

100 **[A1]**

**(b)** Uses trigonometry to find $\angle APB$: $\tan(APB) = \dfrac{60}{80}$ **[M1]**

$\angle APB = \tan^{-1}(0.75) = 36.869...°$ **[A1]**

Bearing $= 040° + 36.869...° = 076.9$ (accept 077) **[A1]**

**(c)** Recognises the shortest distance $d$ is the perpendicular height of $\triangle PAB$ from $A$ to $PB$. **[M1]**

Uses area equation: $\dfrac{1}{2} \times 100 \times d = \dfrac{1}{2} \times 80 \times 60$ OR uses trigonometry $80\sin(53.13°)$ or $60\sin(36.87°)$ **[M1]**

48 **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**4 (a)** Interior angle between parallel North lines $= 180° - 52° = 128°$ **[M1]**

Angles around point $B = 360° - 128° - 142° = 90°$ with no errors seen **[A1]**

**(b)** $AC = \sqrt{15^2 + 20^2}$ **[M1]**

$AC = \sqrt{625}$ **[M1]**

25 **[A1]**

**(c)** $\tan(BAC) = \dfrac{20}{15}$ **[M1]**

Angle $BAC = 53.13°$ **[A1]**

Bearing $= 052° + 53.13°$ **[M1]**

105.1 (or $105.13\ldots$) **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**5 (a)** Sets up equation for $\triangle BCD$: $\tan(42°) = \dfrac{CD}{BC}$

BC $= \dfrac{CD}{\tan(42°)}$ [M1]

Sets up equation for $\triangle ACD$: $\tan(25°) = \dfrac{CD}{BC + 50}$

BC + 50 $= \dfrac{CD}{\tan(25°)}$ [M1]

Equates or substitutes to eliminate $BC$: $50 = \dfrac{CD}{\tan(25°)} - \dfrac{CD}{\tan(42°)}$ **[M1]**

Factorises to solve for $CD$: $CD\left(\dfrac{1}{\tan(25°)} - \dfrac{1}{\tan(42°)}\right) = 50$ **[M1]**

48.4 (accept 48.36...) **[A1]**

**(b)** Recognises the shortest distance $d$ is the perpendicular from $B$ to $AD$, forming a right-angled triangle with hypotenuse $AB = 50$. **[M1]**

Sets up trigonometric ratio: $\sin(25°) = \dfrac{d}{50}$ **[M1]**

21.1 (accept 21.13...) **[A1]**

**[Back to Question]** *[Total Marks: 8]*

**6 (a)** $AC = \sqrt{8^2 + 15^2}$ **[M1]**

17 **[A1]**

**(b)** $AD^2 + 12^2 = 17^2$ (or their $AC^2$) **[M1]**

$AD = \sqrt{289 - 144} = \sqrt{145}$ **[M1]**

12.0 (or $12.04\ldots$) **[A1]**

**(c)** Area of $\triangle ADC = \dfrac{1}{2} \times 12 \times \sqrt{145} = 72.25\ldots$ **[M1]**

$\dfrac{1}{2} \times 17 \times h = 72.25\ldots$ **[M1]**

8.50 (or $8.499\ldots$) **[A1]**

**(d)** $\tan(BAC) = \dfrac{15}{8} \implies$ Angle $BAC = 61.93°$ **[M1]**

$\tan(CAD) = \dfrac{12}{\sqrt{145}} \implies$ Angle $CAD = 44.90°$ **[M1]**

Angle $DAB = 61.93° + 44.90°$ **[M1]**

106.8 (or 106.82 to 106.83) **[A1]**

**[Back to Question]** *[Total Marks: 12]*